AR INFRASTRUCTURE VISUALIZATION SYSTEM

DETAILED RESEARCH DOCUMENT

1. INTRODUCTION

This document explains why an AR Infrastructure Visualization System is needed even when tools like Task Manager exist.

While Task Manager monitors a single local machine, modern infrastructure environments require distributed monitoring systems.

2. WHAT TASK MANAGER DOES

Task Manager is a local operating system utility that:

- Monitors CPU usage
- Monitors memory usage
- Shows running processes
- Displays performance graphs
- Works only on one local machine

Limitations:

- Cannot monitor multiple servers
- No remote access
- No team collaboration
- No infrastructure-wide visualization
- No distributed system monitoring

3. MODERN INFRASTRUCTURE NEEDS

In real-world environments, companies manage:

- Multiple backend servers
- Databases

- Microservices
- Cloud infrastructure
- Load balancers
- Containerized applications

Monitoring such distributed systems requires centralized dashboards.

4. INDUSTRY MONITORING TOOLS

Companies use tools like:

- Prometheus (metrics collection system)
- Grafana (visualization platform)
- Datadog (cloud monitoring)
- New Relic (application performance monitoring)

These tools provide real-time observability across distributed environments.

5. WHAT THIS PROJECT PROVIDES

The AR Infrastructure Visualization System offers:

- Multi-server simulation
- Real-time WebSocket updates
- CPU and memory visualization
- Health status indicators
- Web-based dashboard
- 3D infrastructure representation
- Scalable architecture design

6. REAL-TIME SOCKET COMMUNICATION

Unlike traditional HTTP polling, WebSocket-based communication:

- Maintains persistent connection

- Pushes updates instantly

- Reduces network overhead

- Enables live dashboard updates

7. EDUCATIONAL VALUE

This project teaches:

- Full-stack architecture

- Real-time systems

- Distributed system design

- WebSocket communication

- Git collaboration workflows

- Frontend-backend integration

- System scalability concepts

8. DIFFERENCE BETWEEN LOCAL AND DISTRIBUTED MONITORING

Local Monitoring:

- Single machine

- Manual inspection

- Limited scope

Distributed Monitoring:

- Multiple systems

- Centralized dashboard

- Real-time alerts

- Team visibility

- Scalable infrastructure tracking

9. CAREER & PORTFOLIO VALUE

Building this system demonstrates:

- Engineering problem-solving
- System design capability
- Knowledge of observability concepts
- Real-time data handling
- Modern web architecture skills

10. CONCLUSION

Task Manager is a local troubleshooting tool.

The AR Infrastructure Visualization System is a distributed monitoring platform prototype.

This project simulates real-world observability systems and provides significant educational and professional value.